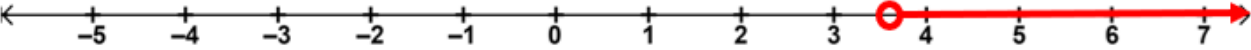
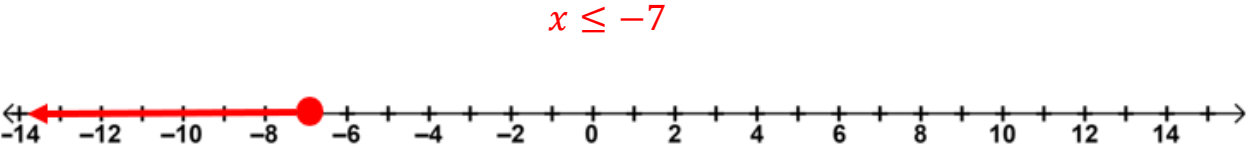


8th Grade Unit 5 Assessment Guide

General Information	At first it may seem that the benchmarks in Unit 5 are unrelated until looking closely at the wording for MA.8.GR.1.3. The first sentence in MA.8.GR.1.3 is, "Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides," so the benchmark MA.8.AR.2.2 is taught first in the unit to support students' understanding of inequalities by solving algebraic inequalities. The second sentence of MA.8.GR.1.3 is, "Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides," so MA.8.GR.1.1 is included to ensure students connect the Pythagorean Theorem and its converse. When students apply the Pythagorean Theorem, students with an understanding of both the Pythagorean Theorem and its converse may choose to solve problems using a variety of methods. Teachers are encouraged to follow student work and to recognize the work of students who can complete steps in their mind. The process taught in Lesson 6 may benefit students who need support, but the assessment does not require the student to apply this process. This is also true for questions for MA.8.GR.1.2. Please note that the distance formula is not taught because the formula is just a modification of the Pythagorean Theorem so students who apply their learning from 6th Grade with MA.6.GR.1.2 using the Pythagorean Theorem as opposed to the distance formula often have a stronger understanding. Many students get confused with which coordinates are which when using the distance formula. Finding the horizontal and the vertical distance and then applying the Pythagorean Theorem alleviates this confusion. The distance formula will be addressed in high school Geometry. In Grade 6, MA.6.GR.1.2, students find distances between ordered pairs that have the same x-coordinate or the same y-coordinate. Teachers may wish to support students who struggle with MA.8.GR.1.2 using resources from Grade 6, Unit 5. In Grade 7, students solve two-step equations (MA.7.AR.2.2) and one-step inequalities (MA.7.AR.2.1). Teachers may wish to support students who struggle with solving two-step inequalities, MA.8.AR.2.2, using resources from Grade 7, Unit 10. The unit also spirals student learning of MA.8.AR.2.3 from Unit 2. When students are solving for sides of triangles, it is important to remind them that the solutions of equations such $x^2 = p$ do result in two values ($\pm\sqrt{p}$), but that in the case of a side length of a triangle the negative solution does not make sense. This also foreshadows student learning of solutions to a quadratic in Algebra 1.
Calculator Information	The questions on the unit assessment are no calculator.
Total Recommended Points	The total recommended points in this guide are 96 points. Teachers can determine how to use the extra 4 points or round the value to 100. Extra points could also be distributed to account for rounding answers with the information provided below in mind. **Rounding Information: The expectations for how students should report their answers when applying the Pythagorean Theorem is left for the teacher to decide. Teachers can determine whether to expect students to report exact answers or rounded values to a specified place value. Answers in the key are rounded to the nearest thousandth, when necessary. Use caution if deducting points for rounding errors as students who make an error rounding may still be demonstrating understanding of the benchmark being assessed.

Question	1	Benchmark(s)	MA.8.AR.2.2
	$x > 3.6$	Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2	Benchmark(s)	MA.8.AR.2.2
	$x \leq -45$	Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	3	Benchmark(s)	MA.8.AR.2.2
	$x < -6.2$	Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	4	Benchmark(s)	MA.8.AR.2.2
	$x \leq 9.8$	Recommended Scoring (7 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	5	Benchmark(s)	MA.8.AR.2.2
	$x > 3.5$ 	Recommended Scoring (10 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider 7 points for the correct solution and 3 points for a correct graph that matches the solution the student found.

Question	6	Benchmark(s)	MA.8.AR.2.2
		Recommended Scoring (10 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider 7 points for the correct solution and 3 points for a correct graph that matches the solution the student found.
Question	7	Benchmark(s)	MA.8.GR.1.1
Vertical Leg 30.544 cm		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	8	Benchmark(s)	MA.8.GR.1.1
Leg 82 units		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	9	Benchmark(s)	MA.8.GR.1.1
Hypotenuse 33.375 yards		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	10	Benchmark(s)	MA.8.GR.1.1
Distance between point <i>K</i> and point <i>L</i> 244.562 ft		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	11	Benchmark(s)	MA.8.GR.1.3
	☐ 58,156,192 ☒ 60,91,109 ☐ 94,168,191 ☒ 115,252,277 ☐ 161,240,290	Recommended Scoring (6 Total Points)	Consider 3 points for each correct answer. Consider deducting 3 points for each incorrect answer up to 6 points.
Question	12	Benchmark(s)	MA.8.GR.1.1
	Length of House 20.616 ft	Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	13	Benchmark(s)	MA.8.GR.1.3
	Possible Side Lengths $41 < x < 153$	Recommended Scoring (6 Total Points)	Consider 3 points for each correct answer.
Question	14	Benchmark(s)	MA.8.GR.1.2
	Distance $\sqrt{221}$ or 14.866	Recommended Scoring (6 Total Points)	Consider 3 points for setting up a correct equation and 3 points for a correct answer.